system with energy

$$\hat{E}(r, \dot{r}) = \frac{\dot{r}^2}{2} - \frac{1}{r} + \frac{l_z^2}{2r^2}.$$

From our previous work, we know that this system (a single degree of freedom system) can be solved for r as a function of t and the initial conditions. Once $r = r(t)$ is known, we may obtain θ by integration:

$$\theta = \theta(0) + \int_0^t \frac{l_z}{r^2(s)} \mathrm{d}s.$$

The example shows that the full solution of the Kepler problem can be worked out given the initial conditions, as long as we are happy to express the solution in terms of antiderivatives of simple functions (and their inverses). Obviously the practical aspects may complicate things, but the idea that the solution can be expressed in this way is already important.

The harmonic oscillator and Kepler problem are both examples of *integrable* systems. In each case there are as many first integrals as there are degrees of freedom. For integrable systems, it is possible, at least locally, to rewrite the equations in new variables that greatly simplify the presentation of the solutions. "Writing a system in new variables" means introducing a change of coordinates. For the harmonic oscillator, recall that the energy is $E(x, v) = v^2/2 + \Omega^2 x^2/2$ and the equations of motion are $\dot{x} = v$, $\dot{v} = -\Omega^2 x$. Define new variables

$$x = \sqrt{2I/\Omega} \cos\theta, \qquad v = \sqrt{2I\Omega} \sin\theta.$$

In these variables, the energy is $E = I\Omega$. Introducing these formulas into the equations of motion and simplifying leads to $\dot{I} = 0$, $\dot{\theta} = -\Omega$. The first equation expresses the constancy of energy; the second describes a rotation with frequency Ω, i.e., the solution is $\theta(t) = \theta(0) - \Omega t$. The variables (I, Ω) are referred to as "action-angle" variables.

A slightly more complicated integrable system would consist of d decoupled harmonic oscillators, each of which has equations of motion $\dot{x}_j = v_j$, $\dot{v}_j = -\Omega_j^2 x_j$. The simple change of variables $((x_j, v_j) \to (I_j, \theta_j))$, applied to each oscillator, would yield equations of motion $\dot{I}_j = 0$, $\dot{\theta}_j = -\Omega_j$. This describes a point winding about a d-dimensional torus defined by angular rotation frequencies Ω_j and radii $|I_j|$ (Fig. 1.14). Depending on the ratio of frequencies such motions may be periodic or quasi-periodic; in the latter case the paths do not "close up" but instead we the curve gradually fills in the surface of the torus. This type of regular motion is sometimes referred to as "tori motion." More generally, one finds occasional examples of nonlinear systems which possesses as many independent first integrals as degrees of freedom (satisfying a certain "involution" condition); such systems may be reduced via coordinate transformation to action-angle variables, i.e. they exhibit tori motion.